Supplementary Information

A Calibrated Transcriptomic Atlas of Porcine Development Reveals Conserved Molecular Programs with Humans

Tianyuan Liu, Rujing Lei, Ilyas M. Khan, Peter Theobald

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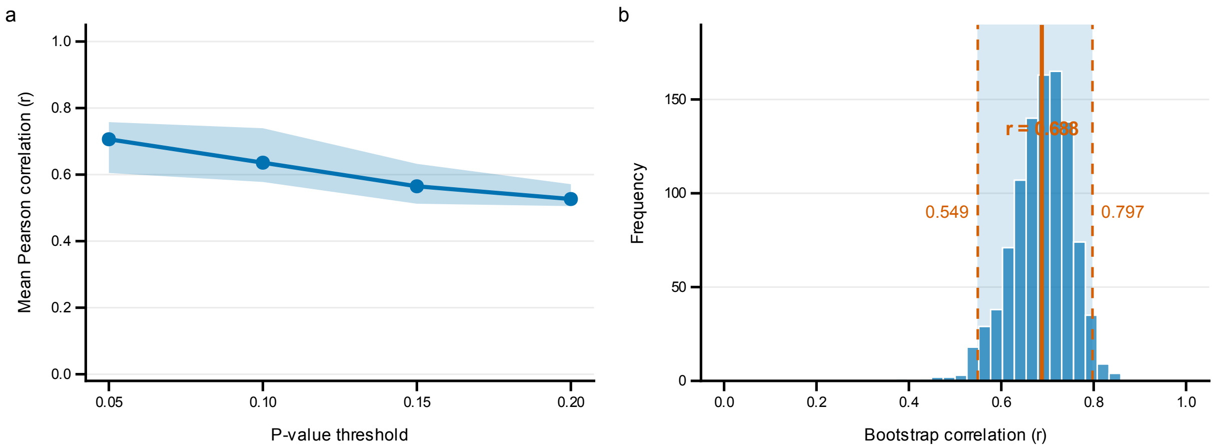
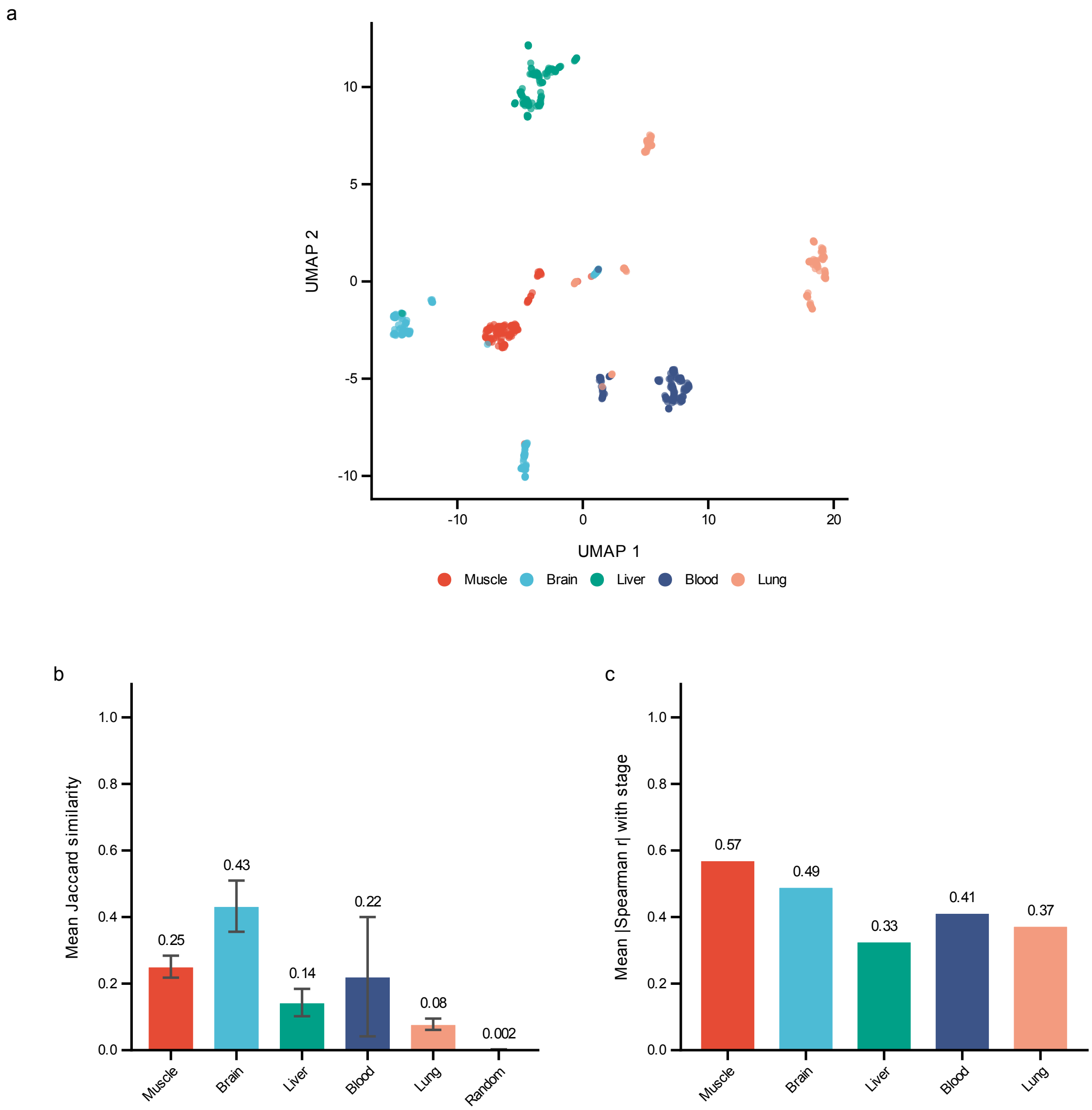
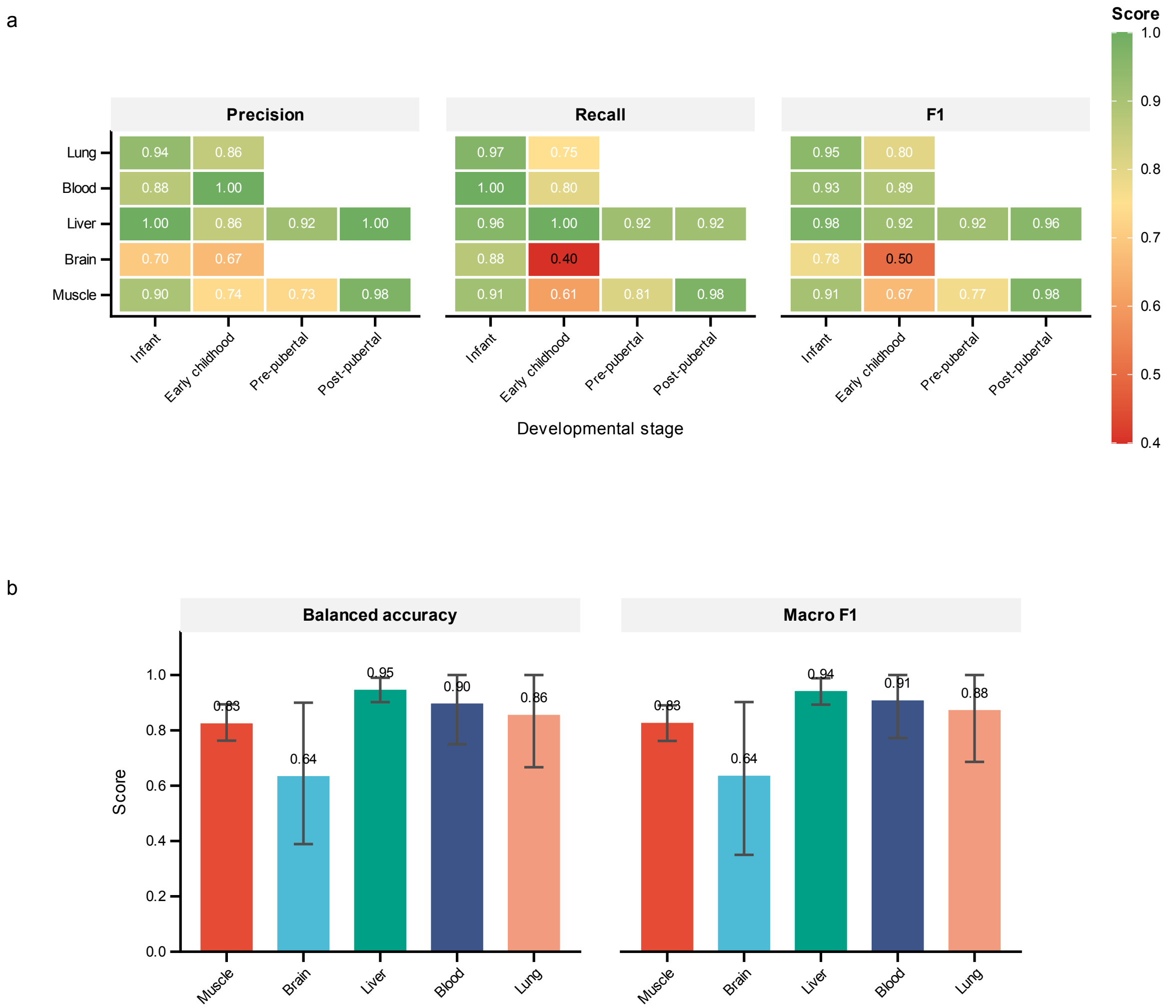


Supplementary Figure 1: Cross-species sensitivity analysis of conserved developmental markers. a, Mean Pearson correlation between pig and human developmental fold changes across significance thresholds ranging from $p < 0.01$ to $p < 0.20$. Shaded region indicates the range (minimum–maximum) across different gene set sizes ($n = 15$–50 genes). **b**, Bootstrap distribution of correlation coefficients ($n = 1{,}000$ iterations) for the optimal parameter combination ($p < 0.05$, $n = 43$ genes). Solid vertical line indicates the observed correlation ($r = 0.685$); dashed lines denote the 95% bootstrap confidence interval boundaries (0.55–0.80).



Supplementary Figure 2: Tissue specificity and feature selection stability analysis. a, Uniform manifold approximation and projection (UMAP) of gene expression profiles across five tissues (muscle, brain, liver, blood, lung), demonstrating distinct tissue-specific clustering patterns ($n = 150$ samples per tissue, balanced sampling; top 500 variable genes). **b**, Within-tissue feature selection stability quantified as mean Jaccard similarity index of the top 50 selected features across ten independent random seeds. Error bars represent standard deviation; higher values indicate more reproducible feature selection. **c**, Mean absolute Spearman correlation between top-ranked features and developmental stage for each tissue.



Supplementary Figure 3: Extended classification performance metrics across tissues and developmental stages. a, Heatmap of per-class precision, recall, and F1 scores for each developmental stage across all five tissues. Colour scale ranges from 0.4 (red, poor performance) to 1.0 (green, excellent performance). Missing values indicate developmental stages not present in that tissue's classification scheme. **b**, Overall performance summary showing balanced accuracy and macro-averaged F1 scores evaluated on the held-out test set (30% of samples) for each tissue. Error bars indicate 95% bootstrap confidence intervals from 1,000 resampling iterations.

Supplementary Tables

Table 1: Train versus test performance comparison. Both train set (70%) and test set (30% held out) performance are reported for all tissues. The gap between train and test metrics indicates the degree of overfitting; regularisation and early stopping were used to minimise this gap.

Tissue	Set	Scheme	n	Balanced Acc.	F1-macro	F1-weighted
Muscle	Train	4-class	761	0.995	0.995	0.996
Muscle	Test	4-class	761	0.828	0.830	0.903
Brain	Train	2-class	43	0.975	0.963	0.967
Brain	Test	2-class	43	0.638	0.639	0.671
Liver	Train	4-class	309	0.987	0.986	0.986
Liver	Test	4-class	309	0.950	0.945	0.947
Blood	Train	2-class	78	0.978	0.981	0.981
Blood	Test	2-class	78	0.900	0.911	0.915
Lung	Train	2-class	129	1.000	1.000	1.000
Lung	Test	2-class	129	0.859	0.876	0.921

Table 2: Per-class performance metrics by tissue. Precision, recall, and F1 scores for each developmental stage within each tissue. n indicates the number of test samples per class.

Tissue	Stage	n	Precision	Recall	F1
Muscle	Infant	58	0.898	0.914	0.906
Muscle	Early childhood	23	0.737	0.609	0.667
Muscle	Pre-pubertal	27	0.733	0.815	0.772
Muscle	Post-pubertal	121	0.975	0.975	0.975
Brain	Infant	8	0.700	0.875	0.778
Brain	Early childhood	5	0.667	0.400	0.500
Liver	Infant	26	1.000	0.962	0.980
Liver	Early childhood	18	0.857	1.000	0.923
Liver	Pre-pubertal	25	0.920	0.920	0.920
Liver	Post-pubertal	24	1.000	0.917	0.957
Blood	Infant	14	0.875	1.000	0.933
Blood	Early childhood	10	1.000	0.800	0.889
Lung	Infant	31	0.938	0.968	0.952
Lung	Early childhood	8	0.857	0.750	0.800

Table 3: Functional annotations of key developmental markers. Annotations compiled from UniProt, GeneCards, and literature review. All listed genes show conserved developmental expression patterns between pig and human muscle tissue. FC = $\log_2$ fold change (adult/infant); positive values indicate higher expression in adults (upregulated during development), negative values indicate higher expression in infants (downregulated during development). Pig FC derived from pig GTEx data (GEO: GSE257558); Human FC from Schaiter *et al. Sci. Rep.* **14**, 22953 (2024).

Gene	Developmental Function	Pig FC	Human FC
SORCS2	Serves as a molecular switch directing trafficking of key signalling receptors to regulate the balance between neuronal growth and pruning, ensuring precise wiring of the developing nervous system.	−1.84	−0.95
FGFRL1	Acts as a "decoy" receptor dampening excessive growth factor signalling to ensure proper formation of critical structures including the diaphragm, kidneys, and skeleton.	−1.68	−2.10
KREMEN1	Functions as a critical negative regulator of Wnt signalling, partnering with Dkk1 to control cell survival and ensure proper patterning of organs including thymus, teeth, and neural tissues.	−1.25	−1.43
ACHE	Functions not only as an enzyme regulating chemical signalling to prevent cell damage but also as a structural adhesion molecule guiding physical growth of neurons and brain wiring.	−1.81	−0.53
DLK1	Crucial regulator of the myogenic program that inhibits myoblast proliferation while promoting differentiation; shows high infant expression that declines as the muscle fiber population matures.	−1.37	−2.88
TRIM54	Muscle-specific E3 ubiquitin ligase (MuRF3) involved in sarcomere Z-disc assembly and organisation; expression increases postnatally as the mature contractile apparatus is established.	1.73	0.36
ME1	Malic enzyme 1, catalysing oxidative decarboxylation of malate to pyruvate; a key source of NADPH for lipid biosynthesis. Expression increases progressively during postnatal development, reflecting metabolic maturation of skeletal muscle.	1.71	0.64

Table 4: Stage-by-stage expression trajectories of key developmental markers. Mean TPM values across porcine developmental stages for genes with conserved pig-human developmental patterns. Stage boundaries: Infant (0–20 d), Early childhood (21–59 d), Pre-pubertal (60–149 d), Post-pubertal (150–365 d), Adult (>365 d). "Peak Change" indicates the developmental transition with the largest fold change.

Gene	Infant	Early	Pre-pub	Post-pub	Adult	Peak Change
SORCS2	14.6	10.9	6.5	0.6	4.1	Pre-pub→Post-pub
FGFRL1	71.2	20.6	27.0	4.3	22.2	Infant→Early
ACHE	16.2	9.9	9.1	1.3	4.6	Pre-pub→Post-pub
KREMEN1	28.9	15.1	12.2	5.2	12.1	Pre-pub→Post-pub
DLK1	301.5	25.8	121.8	0.7	116.5	Infant→Early
TRIM54	68.4	334.5	142.6	377.9	227.5	Infant→Early
ME1	3.1	4.0	3.7	6.9	10.2	Post-pub→Adult